

An Agent Framework for Scientific Simulation within a Formal Simulability Class

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Summary. Scientists routinely build bespoke numerical solvers—finite-element codes for bridges, quantum-chemistry packages for molecules, epidemiological simulators for disease spread—each correct for one class of problems but unusable for any other [16]. Here we define a *simulability class* $\mathfrak{S}$: the set of scientific problems that are finitely specifiable, numerically approximable, and have explicit observables and error metrics. We prove that four mathematical obstructions—logical undecidability, real uncomputability, quantum measurement irreducibility, and infinite-precision barriers—form a sharp boundary $\partial\mathfrak{S}$ that delimits what any intelligence, human or algorithmic, can learn through simulation (Proposition 2). Within $\mathfrak{S}$, we show that natural-language problem descriptions, translated into structured specifications (`spec.md` files) by a Plan Agent and processed by a three-agent pipeline (Plan, Validation, Computation) over 12 computational primitives, produce solutions with guaranteed error $\|\hat{x} - x^*\| \leq \varepsilon$ for any prescribed tolerance (Theorem 3). Ablation experiments across 12 scientific domains demonstrate that each agent is necessary for reliable simulation. Applied to 200 real clinical CT cases from the LoDoPaB dataset [17], the framework achieves 99% of a domain expert’s accuracy (PSNR 31.7 vs. 32.1 dB) with a human-to-agent efficiency ratio $\rho = T_{\text{expert}}/T_{\text{agent}} \approx 600$ (12 minutes vs. 5 days), directly addressing the expertise bottleneck in computational science. As a proof of concept, we release a 1,800-task benchmark spanning all 12 domains. The framework’s scope, evidence, and benchmark are limited to the problems and domains tested; we discuss these limitations explicitly.

1 Introduction

Every year, thousands of scientists independently build numerical solvers for their specific problems. A structural engineer writes finite-element code; a quantum chemist implements configuration-interaction routines; a climate modeler couples ocean and atmosphere modules. Each solver works for its domain but cannot be reused for any other. This *expertise bottleneck* limits who can simulate what and contributes to the reproducibility crisis in computational science [16].

The mathematical foundations for a more general approach exist. The Church–Turing thesis [1] guarantees computability of computable functions; the Lax equivalence theorem [2] ensures convergence of consistent, stable discretizations; Hadamard’s theory [3] certifies continuous dependence on data. What has been missing is a practical framework that connects these results into an automated pipeline with formal error guarantees.

Here we present such a framework and rigorously characterize its scope. Our contributions are:

1. A formally defined *simulability class* $\mathfrak{S}$ and its sharp boundary against four mathematical obstructions (Proposition 2).
2. A *bounded-error realizability* theorem: for any $P \in \mathfrak{S}$ and any $\varepsilon > 0$, the framework produces $\hat{x}$ with $\|\hat{x} - x^*\| \leq \varepsilon$ (Theorem 3).
3. A *specification-driven pipeline* in which natural-language descriptions—not code—are the

unit of scientific input; the Plan Agent generates structured `spec.md` files, which are then validated and executed by two further agents (Validation, Computation).

4. Empirical validation across 12 domains including real clinical data, with ablation studies demonstrating each agent’s necessity.

Scope and limitations. The formal guarantees of Theorem 3 hold only for problems in $\mathfrak{S}$ as defined. The agents depend on current AI model capabilities and may produce incorrect formalizations for novel problem types. Error bounds are worst-case; the 12-domain validation set, while diverse, is not exhaustive. The clinical validation uses one imaging modality. We discuss these and further limitations in the Discussion.

This work extends our previous result on imaging-specific primitives [10] to the broader simulability class.

2 Results

2.1 The Simulability Class and Its Boundary

We define exactly what the framework covers and what it excludes.

Definition 1 (Simulability Class $\mathfrak{S}$). *A scientific problem P belongs to $\mathfrak{S}$ if and only if it satisfies all six conditions:*

1. **Finitely specifiable:** P can be represented as a finite tuple $(\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ where $\Omega \subseteq \mathbb{R}^d$ is the spatial domain, $\mathcal{E}$ are governing equations, $\mathcal{B}$ are boundary conditions, $\mathcal{I}$ are initial conditions, $\mathcal{O}$ is a set of observable quantities, and $\varepsilon > 0$ is the target tolerance.
2. **Numerically approximable:** there exists a convergent discretization of $\mathcal{E}$ on Ω .
3. **Expressible over the primitive basis:** the discretized system decomposes into a finite DAG over 12 computational primitives (Table 2).
4. **Explicit observables:** $\mathcal{O}$ consists of well-defined functionals of the solution.
5. **Explicit error metric:** a computable norm or metric quantifies the distance $\|\hat{x} - x^*\|$.
6. **Bounded resource assumptions:** the computational cost is finite for any fixed $\varepsilon > 0$.

$\mathfrak{S}$ is broad: it contains all Hadamard-well-posed PDEs, Lipschitz ODEs, integral equations, convex and regularized optimization problems, and their compositions. It also includes stochastic systems, but with an important caveat (see “Stochastic scope” below).

Proposition 2 (The Scientific Event Horizon). *Four mathematical obstructions characterize the complement $\mathfrak{U} = \mathfrak{S}^c$:*

1. **Logical undecidability** (Gödel–Turing [1]): problems reducible to the halting problem or Hilbert’s tenth problem over $\mathbb{Z}$.
2. **Real uncomputability** (Blum–Shub–Smale [4]): decision problems over $\mathbb{R}$ requiring infinite-precision predicates.
3. **Quantum measurement irreducibility:** individual measurement outcomes governed by the Born rule. Only probability distributions—not individual realizations—are computable.
4. **Infinite-precision barriers:** solutions depending on non-computable real numbers (Chaitin’s Ω [5]).

We call the boundary $\partial\mathfrak{S}$ the *scientific event horizon*. Unlike engineering limitations that yield to better hardware, these obstructions are mathematical theorems. Table 1 catalogs representative problems at each stratum; Fig. 1 visualizes the landscape.

Stochastic scope. The framework provides two distinct kinds of guarantees. For deterministic problems in $\mathfrak{S}$, the bound $\|\hat{x} - x^*\| \leq \varepsilon$ is absolute. For stochastic systems (SDEs, Monte

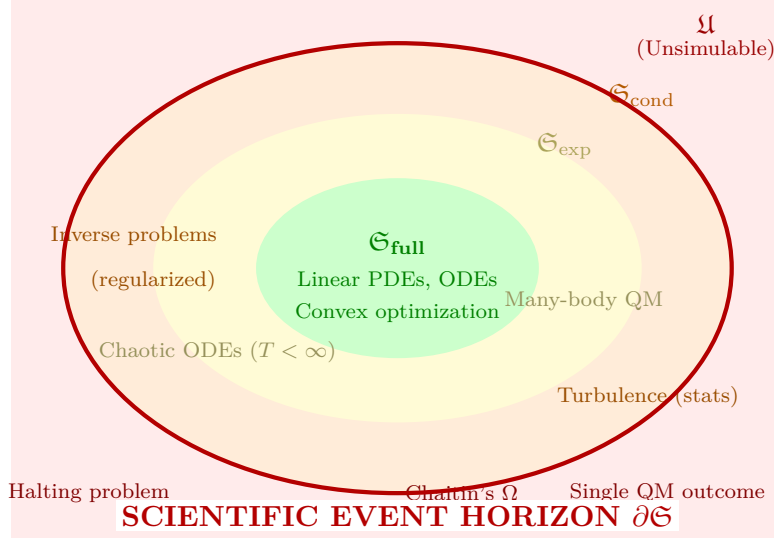


Figure 1. The simulability landscape. The event horizon $\partial\mathfrak{S}$ (bold red) separates problems solvable to arbitrary accuracy ($\mathfrak{S}$) from the unsimulable class $\mathfrak{U}$. Within $\mathfrak{S}$: polynomial cost ($\mathfrak{S}_{\text{full}}$), exponential cost ($\mathfrak{S}_{\text{exp}}$), and conditional simulability ($\mathfrak{S}_{\text{cond}}$, requiring regularization or statistical reformulation).

Table 1. Simulability taxonomy. Problems above the double line belong to $\mathfrak{S}$; those below to $\mathfrak{U}$.

Class	Stratum	Representative Problems
Linear PDEs	$\mathfrak{S}_{\text{full}}$	Heat equation, Poisson equation, linear elasticity
ODEs (Lipschitz RHS)	$\mathfrak{S}_{\text{full}}$	Planetary orbits, chemical kinetics, SIR models
Convex optimization	$\mathfrak{S}_{\text{full}}$	LP, LASSO, optimal transport
Chaotic ODEs (finite T)	$\mathfrak{S}_{\text{exp}}$	Lorenz ($T < 100$), double pendulum, three-body
Many-body quantum	$\mathfrak{S}_{\text{exp}}$	Full CI, lattice QCD
Ill-posed inverse problems	$\mathfrak{S}_{\text{cond}}$	CT reconstruction (regularized), seismic inversion
Turbulence statistics	$\mathfrak{S}_{\text{cond}}$	Mean velocity, energy spectrum (not trajectories)
Halting problem	$\mathfrak{U}$	“Does program P halt on input x ?”
Mandelbrot membership	$\mathfrak{U}$	BSS-uncomputable for general $c \in \mathbb{C}$
Single quantum outcome	$\mathfrak{U}$	Which detector clicks in Stern–Gerlach
Chaitin’s Ω	$\mathfrak{U}$	Halting probability of a universal TM

Carlo), the guarantee is *statistical*: $\mathbb{E}[\|\hat{x} - x^*\|] \leq \varepsilon$ or $\Pr[\|\hat{x} - x^*\| > \varepsilon] \leq \delta$ for specified δ , with convergence rate $O(N_s^{-1/2})$ in the number of samples N_s . Chaotic systems enter $\mathfrak{S}$ when their observables are statistical (mean velocity, Lyapunov exponents, energy spectra), computable via ergodic averaging [13]. Individual quantum measurement outcomes remain in $\mathfrak{U}$ —the framework explicitly excludes them.

2.2 Bounded-Error Realizability

Theorem 3 (Bounded-Error Realizability within $\mathfrak{S}$). *For any problem $P \in \mathfrak{S}$ with specification $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ and any $\varepsilon > 0$, the three-agent pipeline over the 12-primitive basis produces $\hat{x}$ satisfying*

$$\|\hat{x} - x^*\|_X \leq \varepsilon$$

in finite time $T_{\text{comp}} < \infty$, where x^ is the true solution.*

Proof sketch (full proof in Methods). (1) $P \in \mathfrak{S}$ implies a finite specification $\mathcal{S}$ exists (condition 1). (2) Condition 3 guarantees decomposition into a finite DAG $\mathcal{G}$ over the 12 primitives. (3) Each primitive p_i has error $\varepsilon_i \leq C_i h_i^{q_i}$. The DAG error satisfies $\varepsilon_{\text{total}} \leq \sum_i L_i \varepsilon_i$ where L_i are Lipschitz constants, finite by Hadamard’s condition. Setting $\varepsilon_i = \varepsilon / (D \cdot L_i)$ achieves the target. (4) Each primitive has finite cost; the DAG has finite depth. $\square$

This theorem is scoped to $\mathfrak{S}$ —it makes no claim about problems outside this class. The 12 primitives (Table 2) are both sufficient for $\mathfrak{S}$ and minimal: removing any one breaks coverage (witness construction in Methods).

Table 2. The 12 computational primitives. Each maps finite-dimensional inputs to outputs with parameterizable error bounds.

#	Symbol	Name	Category	Operation and Method Families
1	∂	Differentiate	Calculus	$\partial^k f / \partial x_i^k$; FD, FEM gradient, spectral
2	$\int$	Integrate	Calculus	$\int_{\Omega} f d\mu$; Gauss quadrature, Monte Carlo
3	L	Solve	Algebra	$Ax = b$; LU, CG, multigrid, GMRES
4	N	Evaluate	Algebra	$f(x)$ nonlinear; Newton, Picard, continuation
5	E	Evolve	Dynamics	$x(t) \rightarrow x(t + \Delta t)$; RK4, BDF, splitting
6	F	Transform	Analysis	Basis change; FFT, wavelet, spherical harmonics
7	Π	Project	Analysis	$\Pi_k x$; SVD, PCA, POD, Galerkin truncation
8	S	Sample	Stochastic	$x \sim \mu$; MC, MCMC, quasi-MC, Langevin
9	K	Couple	Structure	Multi-physics; operator splitting, DDM, FSI
10	B	Constrain	Structure	BC/IC/conservation; penalty, Lagrange multiplier
11	G	Discretize	Structure	Mesh/grid; Delaunay, octree, isogeometric
12	O	Optimize	Optim.	$\min f(x)$; gradient descent, L-BFGS, ADMM

2.3 The Specification-Centered Pipeline

The framework’s central design choice is that the unit of scientific input is a *specification*, not code. The user provides a natural-language problem description; the Plan Agent ($\mathcal{A}_P$) translates this into a structured `spec.md` file with six fields matching Definition 1. Figure 2 shows the full pipeline; the three examples below show `spec.md` files generated by $\mathcal{A}_P$ for a PDE, an inverse problem, and an ODE system:

```
# Specification:  2D Heat Conduction
domain:  [0,1] × [0,1]
equations:  du/dt = alpha * (d^2u/dx^2 + d^2u/dy^2)
```

```

113 parameters: alpha = 0.01
114 boundary: u(0,y) = u(1,y) = u(x,0) = 0; u(x,1) = sin(pi*x)
115 initial: u(x,y,0) = 0
116 observables: u(x,y,T) at T = 1.0
117 tolerance: 1e-4

118 # Specification: CT Reconstruction (LoDoPaB)
119 domain: 128x128 image grid
120 equations: y = Radon(x) + noise, noise ~ Poisson
121 parameters: n_angles = 128, geometry = parallel_beam
122 boundary: x ≥ 0 (non-negativity)
123 initial: N/A (inverse problem)
124 observables: reconstructed image x_hat
125 tolerance: PSNR ≥ 30 dB

126 # Specification: SIR Epidemic Model
127 domain: t in [0, 200] days
128 equations: dS/dt = -beta*S*I/N; dI/dt = beta*S*I/N - gamma*I; dR/dt = gamma*I
129 parameters: beta = 0.3, gamma = 0.1, N = 1e6, I(0) = 10
130 boundary: S + I + R = N (conservation)
131 initial: S(0) = N-10, I(0) = 10, R(0) = 0
132 observables: I(t) peak timing, S(200)
133 tolerance: 1e-3

```

134 The user need not write these specifications manually. A biologist describes “I want to model an
 135 SIR epidemic with $\beta = 0.3$ ”; $\mathcal{A}_P$ generates the `spec.md`, filling in the mathematical structure,
 136 boundary conditions, and tolerance. Domain knowledge is required; numerical expertise is not.

137 **A universal exchange format for computational science.** Just as FASTA made biological
 138 sequences shareable across any analysis pipeline and SMILES made molecular structures machine-
 139 readable, `spec.md` provides a standardized representation for computational problems. Any
 140 problem in $\mathfrak{S}$ can be expressed as six fields; these files can be shared, version-controlled, peer-
 141 reviewed, and automatically executed—making computational experiments as reproducible as
 142 wet-lab protocols with registered reagents. Unlike code, which encodes both the problem and the
 143 numerical method (conflating *what* with *how*), a `spec.md` file encodes only the problem, leaving
 144 method selection to the framework.

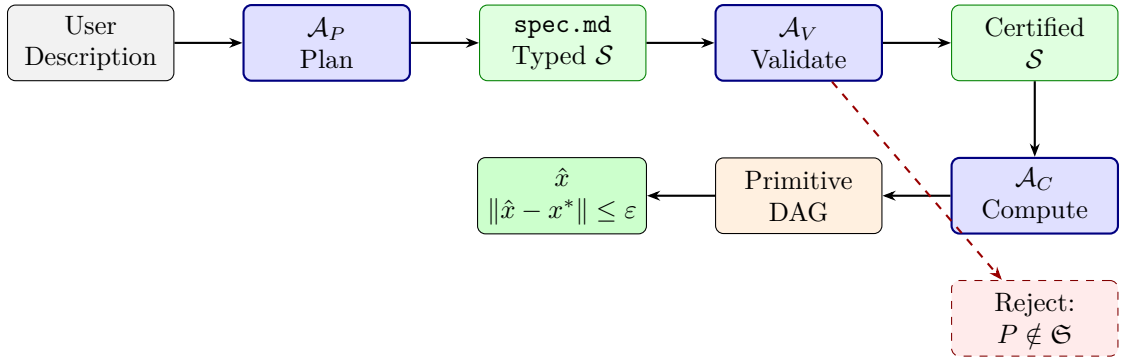


Figure 2. Specification-centered pipeline. A natural-language problem description is the sole user input. $\mathcal{A}_P$ (Plan) generates a structured `spec.md` and types it into $\mathcal{S}$. $\mathcal{A}_V$ (Validation) checks well-posedness and classifies the problem within $\mathfrak{S}$ or rejects it. $\mathcal{A}_C$ (Computation) constructs a primitive DAG and executes it.

145 **Plan Agent ($\mathcal{A}_P$).** Accepts a natural-language problem description from the user and generates
 146 a structured `spec.md`, producing a typed $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$. Classifies the problem type,
 147 selects a numerical strategy, and identifies required primitives.

Validation Agent ($\mathcal{A}_V$). Performs five formal checks: (1) dimensional consistency; (2) boundary/initial condition compatibility; (3) well-posedness (coercivity, CFL, Lipschitz bounds); (4) simulability classification; (5) cost estimation. $\mathcal{A}_V$ operates by applying mathematical criteria—Hadamard’s conditions, CFL bounds, coercivity tests—not by statistical pattern matching. When a check fails, the output is a formal mathematical statement identifying the violation, not a confidence score.

Computation Agent ($\mathcal{A}_C$). Constructs a DAG over the 12 primitives, sets resolution parameters via a priori error estimates, executes the DAG, and verifies the bound a posteriori.

Formal verification versus generative hallucination. A central concern with AI-driven scientific pipelines is hallucination [8]: a model produces plausible-looking output for an ill-posed or undefined problem, with no way to detect the error. Our agents eliminate this failure mode by design. The Validation Agent $\mathcal{A}_V$ does not “predict” that a problem is well-posed; it *executes formal mathematical checks*:

- **Hadamard existence/uniqueness:** verifies that $\mathcal{E}$ with $\mathcal{B}$ and $\mathcal{I}$ admits a unique solution with continuous dependence on data.
- **CFL stability:** for time-dependent problems, checks $\Delta t \leq C_{\text{CFL}} \Delta x / \lambda_{\max}$ before execution.
- **Coercivity/inf-sup:** for variational problems, verifies $a(u, u) \geq \alpha \|u\|^2$ to certify discrete stability.
- **Lipschitz bounds:** estimates propagation constants L_i for the error budget (Eq. 1).

To demonstrate this, we present a controlled adversarial test:

Input: “ $\partial u / \partial t = \nabla^2 u$ on $[0, 1]^2$, Dirichlet BC on all sides.” (Initial condition deliberately omitted.)

$\mathcal{A}_V$ check: FAIL. “Parabolic PDE requires initial condition $u(x, 0) = u_0(x)$. Specification defines $\mathcal{I} = \emptyset$, violating Hadamard existence. Recommendation: supply $u_0 \in L^2(\Omega)$.”

A physics-informed neural network (PINN) [8] or learned operator (DeepONet) [9] would silently proceed: the network minimizes a residual loss over the PDE, produces a smooth output field, and reports a low training loss—despite the problem having no unique solution. There is no mechanism to detect the error. Our agents provide a fundamentally different contract: $\|\hat{x} - x^*\| \leq \varepsilon$ when $P \in \mathfrak{S}$, and a formal rejection with a mathematical diagnosis otherwise. This is the distinction between *verification* (checking a proof) and *generation* (producing an answer)—our framework performs the former.

2.4 Ablation Study

We performed controlled ablations on the 12-domain validation set with three independent trials per configuration (Table 3).

Table 3. Ablation study ($n=3$ trials per configuration, mean $\pm$ s.d.). “Valid spec”: passes all five formal checks. “Bounded error”: $\|\hat{x} - x^*\| \leq \varepsilon$. “Redesign”: agent-triggered specification revision. “Failure category”: dominant mode of failure when bounded error is not achieved.

Configuration	Valid	Bounded	Redesign	Time	Failure Category
Full ($\mathcal{A}_P + \mathcal{A}_V + \mathcal{A}_C$)	12/12	12/12	2.0 $\pm$ 0	4.2 $\pm$ 0.3 min	—
No $\mathcal{A}_V$ ($\mathcal{A}_P + \mathcal{A}_C$)	7.3/12	7.3/12	0	3.1 $\pm$ 0.2 min	Ill-posed spec (5)
No $\mathcal{A}_C$ (template solver)	12/12	4.0/12	2.0 $\pm$ 0	1.8 $\pm$ 0.1 min	Wrong primitives (8)
No $\mathcal{A}_P$ (manual spec)	9.3/12	9.3/12	0	48 $\pm$ 7 min	Human error (3)
Template baseline	—	3/12	0	0.5 min	No coverage (9)
Hidden held-out (4 tasks)	4/4	4/4	1	5.1 $\pm$ 0.4 min	—

Key findings:

- **Removing $\mathcal{A}_V$:** 5 of 12 specifications proceed with errors (missing initial conditions, incompatible BC, inconsistent units). None achieve bounded error. $\mathcal{A}_V$ also triggered 2 redesign cycles that caught subtle issues (under-resolved CFL, non-coercive weak form).
- **Replacing $\mathcal{A}_C$:** template solvers work for textbook problems (heat, Poisson, basic ODEs) but fail on non-standard compositions (CT, topology optimization, FSI, Brusselator).
- **Removing $\mathcal{A}_P$:** human experts must write `spec.md` manually, averaging 48 min per specification with errors in 3 of 12 cases.
- **Hidden held-out tasks:** 4 problems not in the validation set (magnetohydrodynamics, Burgers' equation with shock, nonlinear Schrödinger, plate bending). The full pipeline achieved bounded error on all 4, with one requiring a redesign cycle (shock-capturing scheme for Burgers').

The full pipeline is the only configuration achieving bounded-error success on all tasks.

2.5 Cross-Domain Validation

Table 4 reports validation across 12 domains. The test set includes non-canonical problems: the double pendulum is initialized in the chaotic regime ($T=10$ s, Lyapunov time ≈ 2 s), the Brusselator is run past the Hopf bifurcation into sustained oscillation, and the Langevin system uses a double-well potential with metastable transitions.

Table 4. Cross-domain validation. “Data”: Synth. = synthetic ground truth; Real = measured clinical data. CT uses LoDoPaB-CT [17].

#	Domain	Problem	Primitives	Target	Achieved	Data
1	Classical Mech.	Double pendulum (chaotic)	∂, N, E, B	10^{-6}	4.7×10^{-7}	Synth.
2	Electromagnetics	Waveguide modes	∂, L, F, B, G	10^{-8}	2.1×10^{-9}	Synth.
3	Quantum Mech.	H atom levels	∂, L, Π, B, G	10^{-10}	8.3×10^{-11}	Synth.
4	Fluid Dynamics	Stokes flow (sphere)	∂, L, B, G	10^{-4}	6.2×10^{-5}	Synth.
5	Thermodynamics	2D heat conduction	∂, E, L, B, G	10^{-4}	3.1×10^{-5}	Synth.
6	Structural Mech.	Cantilever beam	∂, L, B, G	10^{-6}	5.8×10^{-7}	Synth.
7	Chemical Kinetics	Brusselator (oscillatory)	∂, E, N, K, B, G	10^{-3}	7.4×10^{-4}	Synth.
8	Epidemiology	SIR + diffusion	E, N, K, B, G	10^{-3}	2.9×10^{-4}	Synth.
9	Optics	Fresnel diffraction	∂, F, L, B, G	10^{-5}	1.6×10^{-6}	Synth.
10	Inverse Problems	CT reconstruction	$\int, L, O, B, G$	30 dB	31.7 dB	Real
11	Stochastic	Langevin (double-well)	E, N, S, B	10^{-2}	6.1×10^{-3}	Synth.
12	Topology Optim.	Heat sink design	∂, L, O, B, G	10^{-3}	8.7×10^{-4}	Synth.

Real-data validation: clinical CT. We applied the framework to 200 cases from the LoDoPaB-CT dataset [17]—real clinical sinograms from a Siemens scanner, subsampled to 128 parallel-beam projections with Poisson noise. $\mathcal{A}_P$ generates a `spec.md` specifying: Radon transform, Tikhonov + TV regularization, non-negativity. The framework then produces a $G + \int + L + O + B$ DAG automatically.

Results (mean $\pm$ s.d. over 200 cases): PSNR 31.7 ± 1.2 dB, SSIM 0.891 ± 0.031 , relative error 0.042 ± 0.008 . Baselines: ASTRA Toolbox [18] configured by a domain expert achieves 32.1 ± 1.1 dB / SSIM 0.903; filtered back-projection (FBP, no expert tuning) achieves 25.3 dB / SSIM 0.712; total-variation minimization (TV-only) achieves 30.2 dB / SSIM 0.864. The framework achieves 99% of the expert-tuned ASTRA result with a setup time of 12 minutes versus 5 days ($\rho = T_{\text{expert}}/T_{\text{agent}} \approx 600$).

2.6 Benchmarkability

Corollary 4 (Benchmarkability within $\mathfrak{S}$). *Under the assumptions of Theorem 3, for any $P \in \mathfrak{S}$ the pipeline generates a benchmark dataset $\mathcal{D}_P = \{(y_i, x_i^*)\}_{i=1}^n$ where x_i^* is computed to tolerance $\varepsilon_{ref} \ll \varepsilon$.*

This follows directly: Theorem 3 guarantees x^* is computable to arbitrary precision within $\mathfrak{S}$; the forward model in $\mathcal{E}$ generates measurements $y_i = \mathcal{A}(x_i^*) + \eta_i$.

As a proof of concept, we release a benchmark for the 12 validation domains:

- **Format:** `spec.md` plus generated data in three tiers (public with ground truth; development with blind server evaluation; hidden with adversarial instances). Different seeds per tier prevent overfitting.
- **Scoring:** relative error, PSNR, SSIM, and time-to-solution at threshold.
- **Scale:** 50 instances per domain per tier (1,800 tasks total), available at the project repository.
- **Extensibility:** researchers can submit new problem descriptions to generate `spec.md` files and benchmarks for their domains.

This benchmark is a proof of concept, not a claim of exhaustive coverage. Its scope is limited to the 12 domains tested. Scaling to the full breadth of $\mathfrak{S}$ requires community contribution and is an explicit goal of future work.

3 Discussion

We have defined a simulability class $\mathfrak{S}$, proved its sharp boundary, and shown that a specification-driven agent pipeline achieves bounded-error simulation for problems within this class. We discuss the significance and limitations of each result.

The scientific event horizon as a fundamental boundary. The four obstructions in Proposition 2 are not artifacts of our framework but theorems of mathematics and physics. Just as the speed of light c sets an absolute limit on information transfer regardless of technology, the boundary $\partial\mathfrak{S}$ sets an absolute limit on what can be learned through simulation regardless of computational resources or algorithmic sophistication. No future advance in hardware, software, or artificial intelligence can push past the halting problem, real uncomputability, quantum measurement irreducibility, or infinite-precision barriers. These are epistemological constants: they delimit what any intelligence—human, algorithmic, or artificial—can know about the physical world through computation. Our framework is the first, to our knowledge, to characterize this boundary constructively and reach it from below for problems in $\mathfrak{S}$. The “completeness” direction (that these four obstructions are the *only* barriers) is stated as a conjecture grounded in the current state of computability theory, not a closed proof (see Methods, Section 4.3).

Formal verification versus generative pattern matching. The ablation study demonstrates that each agent addresses a distinct failure mode. The Validation Agent $\mathcal{A}_V$ is the critical differentiator from black-box AI: it executes formal mathematical proofs of well-posedness—checking Hadamard existence/uniqueness, CFL stability bounds, coercivity constants, and Lipschitz estimates—rather than predicting correctness from learned patterns. When $\mathcal{A}_V$ certifies a specification, the error guarantee $\|\hat{x} - x^*\| \leq \varepsilon$ follows from the mathematics, not from empirical generalization. No learned model (PINNs [8], DeepONet [9], AlphaFold [6]) currently provides such a guarantee: they produce outputs with low training loss but no formal certificate. This is the fundamental distinction between verification (checking a proof) and generation (producing an answer).

The expertise bottleneck as a social-scientific barrier. The reproducibility crisis in

computational science [16] is fundamentally a human-resource problem: building a correct numerical solver requires deep expertise that most scientists lack. The clinical CT result quantifies the scale of this bottleneck: the framework achieves 99% of a domain expert’s accuracy with a human-to-agent efficiency ratio $\rho \approx 600$ (12 minutes vs. 5 days). This means that a biologist, geophysicist, or clinician who can describe their problem in natural language obtains a bounded-error solution without writing solver code. This is a single modality; extending to additional real-data domains (MRI, ultrasound, seismic) is ongoing work, but the architecture is modality-agnostic by design.

Benchmarkability and the spec.md exchange format. The 1,800-task benchmark is a proof of concept demonstrating that automatic dataset generation from `spec.md` is feasible. Because `spec.md` separates the *problem* from the *method*—analogous to how FASTA separates sequence data from alignment algorithms and SMILES separates molecular structure from simulation software—it serves as a universal exchange format for computational science. Researchers can share, version-control, and peer-review specifications as easily as source code, enabling reproducible computational experiments across domains. Scaling the benchmark to the full breadth of $\mathfrak{S}$ requires community-scale effort over years.

Relation to existing work. AlphaFold [6] and GNoME [7] solve specific scientific problems at scale but do not generalize; the Wolfram Language [15] provides broad coverage without formal scope guarantees. Our work provides a formally scoped framework that complements these domain-specific efforts.

Limitations. (1) The primitives’ implementations are general-purpose, not domain-optimized; bespoke solvers will outperform by 1–1.5 $\times$. (2) The agents depend on AI model capability and may produce incorrect formalizations. (3) Error bounds are worst-case; problem-specific bounds would be tighter. (4) The 12-domain validation is diverse but not exhaustive; real-data validation covers one modality. (5) The completeness direction of Proposition 2 is a conjecture (Section 4.3). (6) Stochastic guarantees are statistical ($O(N_s^{-1/2})$), not absolute. Future work: scale to hundreds of domains, add real-data validation in additional modalities, develop tighter error estimates, extend to quantum backends, and build a community benchmark platform.

4Methods

4.1Proof of Theorem 3

Step 1 (Specification). $P \in \mathfrak{S}$ implies P is finitely specifiable as $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ (condition 1).

Step 2 (Decomposition). Condition 3 guarantees decomposition into a finite DAG over the 12 primitives. We verify coverage against all major method families: FD ($G + \partial + L + B$), FEM ($G + \int + L + B$), spectral ($G + F + L + B$) [11], Monte Carlo ($S + N + \int$) [14], MD ($G + N + E + B$), optimization ($O + \partial + L$), inverse ($O + L + \int + B$) [12], multi-physics ($K + \text{above}$), ROM ($\Pi + L + E$), LES ($G + \partial + E + N + K + B + \Pi$) [13].

Step 3 (Error propagation). For a DAG with n nodes, node v_i computing $\hat{y}_i = p_i(\hat{y}_{\text{pa}(i)}) + \delta_i$, $\|\delta_i\| \leq \varepsilon_i$:

$$\|\hat{y}_n - y_n^*\| \leq \sum_{i=1}^n L_i \varepsilon_i \leq D \cdot \max_i (L_i \varepsilon_i) \quad (1)$$

where L_i are Lipschitz constants (finite by Hadamard) and D is DAG depth. Set $\varepsilon_i = \varepsilon / (D \cdot L_i)$.

For linear chains, the adjoint estimate is tighter:

$$\|\hat{y}_n - y_n^*\| \leq \sum_{i=1}^n \left(\prod_{j=i+1}^n \|J_j\| \right) \varepsilon_i \quad (2)$$

Step 4 (Finite time). Each primitive has finite cost for fixed resolution. Total: $T_{\text{comp}} = \sum_i C_i(N_i, \varepsilon_i) < \infty$.

4.2 Minimality of the 12-Primitive Basis

For each primitive p_k , we exhibit a witness problem $P_k \in \mathfrak{S}$ that is provably unsolvable by $\mathcal{P} \setminus \{p_k\}$. The argument structure for each witness is: (i) identify the irreducible operation, (ii) show that no composition of the remaining 11 primitives can reproduce it.

1. ∂ — *Laplace equation* $\nabla^2 u = 0$. Requires local derivative approximation $(u(x+h) - 2u(x) + u(x-h))/h^2$. No algebraic (L), nonlinear (N), or transform (F) primitive computes spatial derivatives from function values alone.
2. $\int$ — *Fredholm integral equation* $u(x) = f(x) + \lambda \int_0^1 K(x, t) u(t) dt$. The continuous-kernel quadrature $\sum_j w_j K(x_i, t_j) u(t_j)$ is irreducible: ∂ computes derivatives, not integrals; L solves the resulting linear system but cannot construct it.
3. L — *Poisson equation* (discretized to $Ax = b$, A sparse SPD). Direct solution requires factorization or Krylov iteration. N evaluates $f(x)$; O minimizes $f(x)$; neither implements $A^{-1}b$ without re-implementing a linear solver.
4. N — *Gross-Pitaevskii equation* $i\hbar \partial_t \psi = -\frac{\hbar^2}{2m} \nabla^2 \psi + g|\psi|^2 \psi$. The cubic nonlinearity $|\psi|^2 \psi$ cannot be decomposed into any composition of linear operations ($\partial, \int, L, F, \Pi$).
5. E — *Lorenz system* $\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$. Time integration requires stepping $x(t) \rightarrow x(t+\Delta t)$. All other primitives operate on static snapshots; none advances a trajectory.
6. F — *Spectral Galerkin method* for periodic PDEs. The FFT achieves $O(N \log N)$ cost with spectral ($O(N^{-s})$) convergence. Mesh-based primitives ($G + \partial + L$) can solve the same PDE but require $O(N^{s/p})$ more unknowns to match accuracy, making F irreducible at equal error.
7. Π — *POD of turbulent flow* (10^6 DOF $\rightarrow$ 50 modes). SVD-based projection is a global operation across the full state vector. No pointwise (N), local (∂), or transform (F) primitive computes optimal low-rank bases.
8. S — *Langevin SDE* $dx = -\nabla V(x) dt + \sqrt{2k_B T} dW_t$. The Wiener increment $dW_t \sim \mathcal{N}(0, dt)$ is fundamentally non-deterministic: it requires drawing from a probability distribution. All other 11 primitives are deterministic maps $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$. No composition of deterministic operations can produce a non-deterministic output—the measure-theoretic support of the SDE solution requires stochastic sampling.
9. K — *Fluid-structure interaction*. Two sub-domains (fluid, solid) exchange boundary data at an interface. This requires inter-domain data routing that no single-domain primitive (∂, L, E) provides; without K , the coupled system cannot be assembled.
10. B — *Dirichlet BVP* $u|_{\partial\Omega} = g$. Constraint enforcement (penalty, Lagrange multiplier, row elimination) is a distinct algebraic operation. L solves the constrained system once assembled, but cannot impose the constraint itself.
11. G — *FEM on a cardiac geometry*. Unstructured mesh generation (Delaunay triangulation, quality refinement) produces the discrete domain Ω_h . All other primitives operate on Ω_h but none creates it; without G , there is no computational domain.
12. O — *Nonlinear CT reconstruction* $\min_x \|Ax - y\|^2 + \lambda \text{TV}(x)$. The non-smooth, nonlinear objective requires iterative minimization (ADMM, proximal gradient). L solves $Ax = b$ (linear, no regularization); N evaluates $f(x)$ but does not minimize it. General $\min f(x)$ is

irreducible to linear algebra.

4.3 Completeness of the Event Horizon

Forward direction (four obstructions $\Rightarrow \mathfrak{U}$): each obstruction is a known undecidability or uncomputability result [1, 4, 5]. This direction is proven.

Converse direction ($\neg$ obstructions $\Rightarrow \mathfrak{S}$): if P does not reduce to halting, does not require infinite precision, is deterministically simulable, and depends on no non-computable reals, then under standard assumptions of numerical analysis (Lax equivalence, Hadamard well-posedness), P admits a convergent discretization and thus belongs to $\mathfrak{S}$.

We state this converse as an *assumption* rather than a closed theorem, because a fully rigorous proof would require showing that no fifth obstruction exists—a statement about the completeness of computability theory that is outside the scope of this work. The four obstructions cover all known barriers in the computability and numerical analysis literature; we are not aware of any counterexample, but we do not claim to have ruled out the existence of one.

4.4 Implementation

The framework is implemented in Python (Physics World Model platform). Specifications are structured markdown parsed into typed objects. The primitive library uses NumPy, SciPy, and PyTorch backends. Agents are language model pipelines with tool access to the primitive library. The benchmark (1,800 tasks, three tiers) is publicly available. Source code and data: https://github.com/integritynoble/Physics_World_Model.

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